



Cairo University



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Statement of Work (SOW) for Outpatient Clinics Management System

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SOW for Clinic Patient Management System

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1. Introduction and Background

1.1 Introduction

The project focuses on designing and implementing a mobile/web Outpatient Clinic Management System for Kasr Al Ainy Hospital, one of Egypt's most prestigious and historic medical institutions. Kasr Al Ainy Hospital serves a vast number of citizens daily, providing essential healthcare services. However, inefficiencies in patient data registration and tracking hinder the delivery of optimal care. Our solution aims to address these challenges by leveraging modern technologies to streamline data collection, storage, and retrieval, ultimately enhancing healthcare quality and patient experience in the outpatient clinics

1.2 Background

Kasr Al Ainy Hospital's outpatient clinics handle a high volume of patients, making efficient data management critical for effective treatment and follow-up care. Currently, the process of recording patient information, such as name, age, national ID, medical history, diagnoses, lab results, and imaging reports, relies on manual methods that are prone to errors, loss of records, delays, and sometimes this data is not recorded. This leads to fragmented patient histories, repeated data entry during subsequent visits, and reduced overall efficiency for both doctors and patients. The absence of a centralized, digital system results in suboptimal resource utilization and potential gaps in continuity of care, which can impact patient outcomes in a facility that plays a pivotal role in Egypt's public health system.

The proposed solution emerges from the need to modernize these processes in a resource-constrained environment. Recognizing the limited availability of computers in every clinic and the ubiquity of mobile devices among physicians, our team has conceptualized a mobile/web application that integrates seamlessly with hospital-provided servers. This approach ensures accessibility, scalability, and security while aligning with the hospital's infrastructure

1.3 Proposed Solution

The Outpatient Clinic Management System will be a mobile/web application designed for Android and iOS devices, enabling physicians in Kasr Al Ainy's outpatient clinics to efficiently collect, store, and access patient data in more than 50 different clinics.

2. Modules and Features

The system is divided into key modules: Admin (Super User), Doctor, Patient, GenAI Chatbot, AI Enhanced EHR, and Data Entry. Each module includes specific features tailored to the project's requirements.

2.1 Admin Module (Super User)

This module is accessible only to super users for high-level system management.

Features:

- a. **Add new clinics:** Create clinic profiles with details like name, location, and associated doctors.
- b. **Delete clinics:** Remove inactive clinics, with confirmation prompts to prevent data loss.
- c. **Add User:** Create doctors, patients.
- d. **Remove User:** Remove doctors, patients.
- e. **Overall system monitoring:** System analytics dashboard.
- f. **QR code generation:** Create patient cards with QR codes linked to National ID, enabling navigation to offline-stored patient history. (accessible only for EL Kasr El Ainy Doctors)

2.2 Doctor Module

Doctors are the primary system users, with full access to patient-related functionalities.

Features:

- a. **Patient search and profile management:** Search by National Number, Name, or other identifiers, view existing patient history or create new profiles if none exist.
- b. **Edit patient information:** Update demographics (Date of Birth, Gender, National Number, Name) and medical details at any time.
- c. **Patient history access and updates:** View and add to history sections including labs (e.g., blood test results), diagnoses (e.g., hypertension), treatment plans (e.g., medication regimens), doctor's notes, visit dates, and clinical status (e.g., stable/improving).

2.3 Patient Module

Patients are the second most important users in the system. The system should make their cycle easier, faster, and safer

Features:

- a. **Patient Profile:** His past visits dates, and results (Medications, Diagnoses, Cost), His upcoming appointments, and His EMR
- b. **Booking and Appointment scheduling:** patients can choose a clinic and see all the available doctors in this clinic, and schedule an appointment through the system
- c. **Medications:** The system should automatically notify (Email, Push Notification) the patient of his medication times
- d. **Online payment:** The patient can choose whether to pay online or offline for his/her appointments, and the system should allow him/her to pay securely if he/her choosed to pay online (Integration with a payment gateway)

2.4 GenAI Chatbot

Helping customer relations functions by replacing FAQs and similar general stuff

2.5 AI Enhanced EHR

Analyzing the patient history and aiding doctors in diagnoses

Features:

- a. **Automated Summarization:** Reads a patient's entire (and often lengthy) chart and generate a concise summary of their medical history, recent lab results, and active issues, preparing the clinician for the visit in seconds.
- b. **Diagnostic Assistance:** Analyzes medical images (like X-rays, CT scans, or MRIs) and highlight potential abnormalities, such as tumors or fractures, for the radiologist to review. It can also suggest potential diagnoses based on a patient's symptoms and lab results.
- c. **Medication Safety:** The system can perform sophisticated checks for potential drug-to-drug or drug-gene interactions, alerting the provider to potential adverse effects based on the patient's unique genetic profile and medical history.

2.6 Data Entry Module

Central to the system, this module handles input via ML technologies, ensuring efficient and accurate data capture.

Features:

- a. **Speech-to-Text:** Doctors dictate entries in Arabic or English, and the system transcribes in real-time with editable output (e.g., correct misrecognized medical terms like "dyspnea").
- b. **Optical Character Recognition (OCR):** Doctors can scan patients' ID cards for automatic text extraction and population into patient fields.
- c. **Bilingual support:** Handle mixed-language inputs seamlessly.
- d. **Validation and editing:** Post-transcription review interface to edit text before saving.
- e. **Integration:** All entries feed into the patient history.

2.7 Non-Functional Requirements

An essential non-functional requirements including:

- a. **Offline Access:** enabling offline operations like accessing patients' history, and schedules to overcome disconnectivity and system failures
- b. **Scalability:** Support scalability with up to 50 clinics and future expansions.
- c. **Applying Cloud Native Software Engineering:** Microservices architecture, Containerization for each service, Building CI/CD pipelines, DevOps, and MLOps
- d. **Applying Automation/Manual Testing Techniques:** Unit testing, Integration testing, and E2E testing
- e. **Data Encryption:** Encrypting all patients' data following the global standards for data privacy and security

3. Deliverables

- a. **Prototype Application:** Fully functional web/mobile app with all modules implemented.
- b. **Documentation:** User manual, technical architecture diagram, code repository (e.g., GitHub).
- c. **Demo and Presentation:** Live demonstration of key features.
- d. **Source Code:** Clean, commented code with setup instructions.
- e. **Testing Reports:** Bug logs and resolution summaries.